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Open Geospatial Consortium

BEST PRACTICE FOR OGC GEODATACUBE MANAGEMENT

BEST PRACTICE
General

CANDIDATE SWG DRAFT

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Note

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CONTENTS

I ABSTRACT

The rapidly growing volume of Earth observation data, together with the increasing diversity of data providers, platforms, formats, APIs, and processing environments, makes it difficult for users to consistently discover, access, prepare, and use data for analysis. While metadata standards and catalogues provide an essential foundation for discovery and description, additional guidance is needed on how discovered datasets can be transformed into analysis-ready geodatacubes and managed throughout their use.

This Best Practice provides recommendations for geodatacube management, focusing on the pathways from metadata and data discovery towards usable multidimensional datacubes. It describes how metadata-driven approaches can support the creation, configuration, access, and processing of geodatacubes across heterogeneous infrastructures. Building on previous work, including OGC Testbed 19, the document explores the role of openEO and relevant open-source EOEPCA building blocks as interoperable components for exposing, processing, and managing Earth observation data as geodatacubes.

The goal of this Best Practice is to provide practical guidance for data providers, platform operators, and application developers who need to design geodatacube management systems. It recommends approaches for bridging discovery metadata, data access mechanisms, processing interfaces, and datacube representations in a way that improves interoperability, reproducibility, and usability of Earth observation data.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, best practice, GeoDataCubes

III PREFACE

NOTE: Insert Preface Text here. Give OGC specific commentary: describe the technical content, reason for document, history of the document and precursors, and plans for future work.

There are two ways to specify the Preface: “simple clause” or “full clasuse”

If the Preface does not contain subclauses, it is considered a simple preface clause. This one is entered as text after the `.Preface` label and must be placed between the AsciiDoc document attributes and the first AsciiDoc section title. It should not be give a section title of its own.

If the Preface contains subclauses, it needs to be encoded as a full preface clause. This one is recognized as a full Metanorma AsciiDoc section with te title “Preface”, i.e. `== Preface`. (Simple preface content can also be encoded like full preface.)

IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- Eurac Research
- Organization two
- Organization three

VI SUBMITTERS

All questions regarding this submission should be directed to the editor or the submitters:

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VII CONTRIBUTORS

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NOTE: If you need to place any further sections in the preface area use the `[.preface]` attribute.

1

SCOPE

1 SCOPE

NOTE: This document specifies a set of best practice recommendations for the management of geospatial data as geodatacubes. The scope of this document is the transition from metadata-driven discovery of Earth observation datasets to their access, organization, processing, and use as multidimensional geodatacubes.

The document provides guidance on the use of metadata records, catalogues, data access information, and processing interfaces to support interoperable geodatacube management. It includes consideration of openEO as an interface for geodatacube access and processing, and considers the use of open-source EOEPCA building blocks as components within geodatacube management architectures.

This document is intended to support implementers of data platforms, processing services, and client applications that need to manage Earth observation data in a datacube-oriented form. It does not define a new standard for geodatacube encoding, metadata, or processing. Instead, it identifies recommended practices for combining existing standards, interfaces, and software components to support consistent and reusable geodatacube workflows.

2

CONFORMANCE

2 CONFORMANCE

This Best Practice defines XXXX.

Requirements for N standardization target types are considered:

- AAAA
- BBBB

Conformance with this Best Practice shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® interface Best Practice, a software implementation shall choose to implement:

- Any one of the conformance levels specified in Annex A (normative).
- Any one of the Distributed Computing Platform profiles specified in Annexes TBD through TBD (normative).

All requirements-classes and conformance-classes described in this document are owned by the documents(s) identified.

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

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4

TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. example term

term used for exemplary purposes

Note 1 to entry: An example note.

Example Here’s an example of an example term.

[**SOURCE:** ISO 19101-1:2014]

5

CONVENTIONS

5 CONVENTIONS

This sections provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

5.1. Identifiers

The normative provisions in this document are denoted by the URI

`http://www.opengis.net/spec/{standard}/{m.n}`

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

6

CLAUSES NOT CONTAINING NORMATIVE MATERIAL

6 CLAUSES NOT CONTAINING NORMATIVE MATERIAL

Paragraph

6.1. Clauses not containing normative material sub-clause 1

Paragraph

6.2. Clauses not containing normative material sub-clause 2

7

CLAUSE CONTAINING NORMATIVE MATERIAL

7 CLAUSE CONTAINING NORMATIVE MATERIAL

Paragraph

7.1. Requirement Class A or Requirement A Example

Paragraph – intro text for the requirement class.

Use the following table for Requirements Classes.

REQUIREMENTS CLASS 1	
OBLIGATION	requirement
DESCRIPTION	Requirements Class • http://www.example.org/req/blah • urn:iso:ts:iso:19139:clause:6
NORMATIVE STATEMENTS	Requirement 1-1 requirement description Requirement 1-2 requirement description Requirement 1-3 requirement description

7.1.1. Requirement 1

Paragraph — intro text for the requirement.

Use the following table for Requirements, number sequentially.

REQUIREMENT 1	
OBLIGATION	requirement
STATEMENT	Requirement ‘shall’ statement

Dictionary tables for requirements can be added as necessary. Modify the following example as needed.

Table 1

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
name 1	definition of name 1	float	One or more (mandatory)
name 2	definition of name 2	character string type, not empty	Zero or one (optional)
name 3	definition of name 3	GML:: Point PropertyType	One (mandatory)

A

ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

A ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

NOTE: Ensure that there is a conformance class for each requirements class and a test for each requirement (identified by requirement name and number)

A.1. Conformance Class A

A.1.1. Requirement 1

REQUIREMENT A.1	
TEST PURPOSE	Verify that...
TEST METHOD	Inspect...

A.1.2. Requirement 2

B

ANNEX B (INFORMATIVE) TITLE ({NORMATIVE/INFORMATIVE})

B

ANNEX B (INFORMATIVE) TITLE ({NORMATIVE/INFORMATIVE})

NOTE: Place other Annex material in sequential annexes beginning with “B” and leave final two annexes for the Revision History and Bibliography

C

ANNEX C (INFORMATIVE) REVISION HISTORY

C

ANNEX C (INFORMATIVE) REVISION HISTORY

Table C.1

DATE	RELEASE	EDITOR	PRIMARY CLAUSES MODIFIED	DESCRIPTION
2016-04-28	0.1	G. Editor	all	initial version

BIBLIOGRAPHY

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